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Last time. $G \curvearrowright X$ 平移一般不是自同态，但共轭是自同构

$G \overset{Ad}{\curvearrowright} G$ → 保持群结构

$Ad: G \xrightarrow{\text{hom.}} Aut(G)$

$g \mapsto Ad_g \quad G \xrightarrow{\sim} G,\ x \mapsto gxg^{-1}$

$N \rtimes_\phi H$

§6. More on group actions: Stablizers, orbits, class equations, outer automorphisms

DEF Let G be a group acting on a set X $G \curvearrowright X$.

For each $x \in X$, $Stab_G(x) := \{g \in G \mid g \cdot x = x\} < G$ called the stablizer subgp (稳定子群) of x.

$Orb_G(x) := G \cdot x = \{g \cdot x \mid g \in G\} \subset X$ called the orbit (轨道) of x

Properties.

1. $Stab_G(x)$ is a subgp of G

$(\forall g, h \in Stab_G(x), (gh^{-1}) \cdot x = g \cdot (h^{-1} \cdot x) = g \cdot (h^{-1} \cdot (h \cdot x)) = g \cdot x = x \Rightarrow gh^{-1} \in Stab_G(x))$.

2. $\forall x, y \in X$, either $Orb_G(x) = Orb_G(y)$

or $Orb_G(x) \cap Orb_G(y) = \varnothing$

$\Rightarrow X = \bigsqcup$ Orbits

(if $z \in Orb_G(x) \cap Orb_G(y) \Rightarrow z = gx = hy$ for some $g, h \in G$.

$\Rightarrow \forall w \in Orb_G(x)$, $w = kx$ for some $k \in G$

$w = kg^{-1}z = (kg^{-1}h)y \in Orb_G(y)$ and vice versa $\Rightarrow Orb_G(x) = Orb_G(y)$)

补：同一轨道上的稳定子彼此共轭.

若 $y = g \cdot x$, 则 $Stab_G(y) = g\, Stab_G(x)\, g^{-1}$.

证：$h \in Stab_G(y) \Leftrightarrow h \cdot y = y \Leftrightarrow h \cdot (g \cdot x) = g \cdot x \Leftrightarrow (g^{-1}hg) \cdot x = x$

$\Leftrightarrow g^{-1}hg \in Stab_G(x) \Leftrightarrow h \in g\, Stab_G(x)\, g^{-1}$. □

Important case. $G \overset{Ad}{\curvearrowright} G$.

Recall Two elements $a, b \in G$ are conjugate if $a = gbg^{-1}$ for some $g \in G$.

Conjugacy classes = orbits of the conjugation action $G \overset{Ad}{\curvearrowright} G$.

eg. if G is abelian, the conjugation of a is $\{a\}$.

eg. $G = GL_n(\mathbb{C})$ conjugation classes $\overset{1-1}{\longleftrightarrow}$ Jordan forms (with nonzero eigenvalues and up to permutations) (一一对应)

eg. $G = S_n$ conjugation classes $\overset{1-1}{\longleftrightarrow}$ partitions of n → 将n写成正整数之和的方法数，即置换的型 (不计顺序)

(Recall for $\sigma\in S_n$, $\sigma(a_1\cdots a_r)\sigma^{-1}=(\sigma(a_1)\cdots\sigma(a_r))$ conjugation keeps the cycle decomposition type.

On the other hand, if τ_1,τ_2 have the same cycle type, then $\exists\sigma\in S_n$, st. $\sigma\tau_1\sigma^{-1}=\tau_2$.)

DEF Let $H<G$ be a subgp. $S\subset G$ a subset.

1). $C_G(S):=\{g\in G\mid \forall s\in S,\ gsg^{-1}=s\}$ $(gs=sg)$

$=\bigcap_{s\in S}\mathrm{Stab}_G(s)<G$ → for the conjugation action

called the centralizer (中心化子) of S. When $S=\{s\}$, also write $C_G(s)$ $=\mathrm{Stab}_G(s)$ 在共轭作用下

2). $Z(G):=\{g\in G\mid \forall h\in G,\ ghg^{-1}=h\}$ $(gh=hg)$

$=C_G(G)<G$ called the center (中心) of G.

alternatively $Ad: G\to \mathrm{Aut}(G)$ Then $Z(G)=\ker(Ad)\lhd G$. (later about $\mathrm{Im}(Ad)$)

3). $N_G(H):=\{g\in G\mid gHg^{-1}=H\}$ (as sets) called the normalizer (正规化子) of H.

(in particular, if $H\lhd G$, then $N_G(H)=G$)

alternatively, $G\curvearrowright\{\text{all subgps of } G\}^{\supseteq H}$ $g\cdot H=gHg^{-1}$

Then $N_G(H)=\mathrm{Stab}_G(H)<G$.

$g\in C_G(H)$: g 的共轭逐个固定 H 中每个元素

$g\in N_G(H)$: g 的共轭固定 H (作为整体)

} $C_G(H)<N_G(H)$

$H\lhd N_G(H)$ $H\lhd G\Leftrightarrow N_G(H)=G$.

$N_G(H)$ 即为 G 中使 H 成为正规子群的最大子群.

DEF Let $G\curvearrowright X$ $G\curvearrowright Y$

we say a map $\Phi: X\to Y$ is G-equivariant (G-等变)

if $\forall g\in G, x\in X$, $\Phi(g\cdot x)=g\cdot\Phi(x)$.

每引入一种结构,通常要带着这个结构研究与之相容的映射.

REM vector spaces ~ linear maps 保持线性结构

groups ~ homomorphisms 保持乘法

"G-sets" sets with G-actions ~ G-equiv maps 保持群作用结构

DEF Let $G\curvearrowright X$. We say the action is transitive (传递)

if $\forall x,y\in X$, $\exists g\in G$, st. $gx=y$ $(\Leftrightarrow X=\text{single orbit})$

固定 x, 考虑最自然的映射 $G\to \mathrm{Orb}_G(x)$, $g\mapsto gx$ 显然满但不单

$g_1x=g_2x\Leftrightarrow(g_1^{-1}g_2)x=x\Leftrightarrow g_1^{-1}g_2\in\mathrm{Stab}_G(x)$

$\Leftrightarrow g_1\mathrm{Stab}_G(x)=g_2\mathrm{Stab}_G(x)$

in this case $\Phi: G/\mathrm{Stab}_G(x)\to X$ (未必是商群) is a G-equivariant bijection.

$G/\mathrm{Stab}_G(x)$: set of left cosets; G acts by left translation

$g\mathrm{Stab}_G(x)\mapsto gx$

自然要考虑 $G/\mathrm{Stab}_G(x)\to\mathrm{Orb}_G(x)$

(Set $H = Stab_G(x)$

Φ is well-defined: if $g_1H = g_2H \Rightarrow g_1 = g_2h \Rightarrow g_1x = g_2hx = g_2x$

surjective: obvious.

injective: if $\Phi(g_1H) = \Phi(g_2H) \Rightarrow g_1x = g_2x \Rightarrow g_1^{-1}g_2x = x \Rightarrow g_1^{-1}g_2 \in H \Rightarrow g_1H = g_2H$.

G-equivariant: $\Phi(g' \cdot gH) = \Phi(g'gH) = g'gx = g' \cdot \Phi(gH)$)

More generally $X = \bigsqcup$ orbits.

作为带G-作用的集合(两边只是一般的集合,未必是群)

$\forall x \in X$, G acts transitively on $Orb_G(x) \Rightarrow Orb_G(x) \simeq G/Stab_G(x)$

$\Rightarrow X \simeq \bigsqcup_{\text{orbits } Orb_G(x)} G/Stab_G(x)$ 按轨道拆的集合 轨道事实上是一个陪集空间

bijection; one representative for each orbit

THM Let G be a finite group (acting on itself by conjugation) $Ad: G \to Aut(G)$, $h \mapsto Ad_h: G \xrightarrow{\sim} G$, $g \mapsto hgh^{-1}$

1) $\forall g \in G$, the number of elements in its conjugacy class is

$\#(Ad_G(g)) = \#G / \#C_G(g) = [G : C_G(g)]$.

2) class equation (类方程): if $g_1, \cdots, g_r$ are representative of the conjugacy classes of G

Then $\#G = \sum_{i=1}^{r} [G : C_G(g_i)]$.

REM Conjugacy class $Ad_G(g)$ is a singleton $\{g\} \Leftrightarrow \forall h \in G, hgh^{-1} = g \Leftrightarrow g \in Z(G)$.

$\Rightarrow \#G = \#Z(G) + \sum_{\substack{\text{nontrivial conjugacy classes} \\ Ad_G(g_i)}} [G : C_G(g_i)]$

DEF Let p be a prime number.

A finite group G is called a p-group if $\#G = p^r$ for some r.

PROP For a nontrivial p-group G, $Z(G)$ is nontrivial.

$\left(\#G = \#Z(G) + \sum [G : C_G(g_i)]\right)$

nontrivial p-power; nontrivial p-power

Recall $Ad: G \to Aut(G)$

$\ker(Ad) = Z(G)$

$\mathrm{im}(Ad) =: \mathrm{Inn}(G) < \mathrm{Aut}(G)$ is called the group of <u>inner automorphism</u> (内自同构) of G.

PROP $\mathrm{Inn}(G) \triangleleft \mathrm{Aut}(G)$.

Pf Want: $\forall \sigma: G \to G$ automorphism $\quad \sigma \mathrm{Inn}(G) \sigma^{-1} = \mathrm{Inn}(G)$ (suffices to show $\subset$)

Take any $g \in G \rightsquigarrow Ad_g: G \xrightarrow{\sim} G$, $x \mapsto gxg^{-1}$

Compute $\sigma Ad_g \sigma^{-1}: G \xrightarrow{\sim} G \qquad \sigma Ad_g \sigma^{-1}(x) = \sigma(g\sigma^{-1}(x)g^{-1}) = \sigma(g) x \sigma(g)^{-1} = Ad_{\sigma(g)}(x)$

$\Rightarrow \sigma Ad_g \sigma^{-1} = Ad_{\sigma(g)} \in \mathrm{Inn}(G) \Rightarrow \mathrm{Inn}(G) \triangleleft \mathrm{Aut}(G)$. □

DEF The $\mathrm{Out}(G) := \mathrm{Aut}(G)/\mathrm{Inn}(G)$ is called the group of <u>outer automorphism</u> (外自同构) of G.

eg. $G = S_n$, $n \geq 3$. $Ad: S_n \to \mathrm{Aut}(S_n)$ injective → 因为核(即$Z(S_n)$)是平凡的

interesting fact: if $n \neq 6$, $Ad: S_n \xrightarrow[\text{isomorphism}]{\sim} \mathrm{Aut}(S_n)$ "all automorphisms are inner automorphisms".

But when $n=6$, $\exists \Phi: S_6 \xrightarrow{\sim} S_6$, $(12) \mapsto (12)(34)(56)$ that is not inner.

in fact, $\mathrm{Aut}(S_6) \simeq \mathrm{Inn}(S_6) \rtimes \{1, \Phi\}$

Construction of Φ: $PGL_2(\mathbb{F}_5) \curvearrowright \mathbb{P}^1(\mathbb{F}_5)$ faithful & transitive

$\#PGL_2(\mathbb{F}_5) = 120 \qquad \#\mathbb{P}^1(\mathbb{F}_5) = 6$.

$\rightsquigarrow$ get $PGL_2(\mathbb{F}_5) \hookrightarrow S_{\mathbb{P}^1(\mathbb{F}_5)} \simeq S_6 \qquad \#(S_6/PGL_2(\mathbb{F}_5)) = \frac{720}{120} = 6$.

Consider $S_6 \curvearrowright S_6/PGL_2(\mathbb{F}_5) =: X$ action by left translation

$\rightsquigarrow$ get $\Phi: S_6 \to S_X \simeq S_6$ homomorphism

S_6的左陪子群

$\ker\Phi = \{1\}, A_6, S_6$ (360, 720)

but if $g \in \ker\Phi$, $gPGL_2(\mathbb{F}_5) = PGL_2(\mathbb{F}_5) \Rightarrow g \in PGL_2(\mathbb{F}_5)$ (120) $\Rightarrow \ker\Phi = \{1\} \Rightarrow \Phi \in \mathrm{Aut}(S_6)$.

(secretly, $PGL_2(\mathbb{F}_5) \simeq S_5 \qquad PSL_2(\mathbb{F}_5) \simeq A_5$

$\rightsquigarrow$ get an exotic embedding $S_5 \hookrightarrow S_6$)

DEF Let G be a group. We say a subgroup $H < G$ is characteristic (denote $H \mathrm{ char } G$)

if for any automorphism σ of G, $\sigma(H) = H$.

Properties

1). characteristic $\Rightarrow$ normal

2). if $H<G$ is the unique subgp of G of that order $\Rightarrow H$ char G.

eg. in $\mathbb{Z}/n\mathbb{Z}$ $\forall d|n$, $\langle d\rangle$ char $\mathbb{Z}/n\mathbb{Z}$.

3). K char H, $H\triangleleft G \Rightarrow K\triangleleft G$.

K char H, H char $G \Rightarrow K$ char G 注意$\triangleleft$不具有传递性.

! K char G, $K<H<G \not\Rightarrow K$ char H 因为H可能有比G更"多"的自同构. 对比 $K\triangleleft G$, $K<H<G \Rightarrow K\triangleleft H$.

($\forall g\in G$, $gHg^{-1}=H \Rightarrow Ad_g: H\xrightarrow{\sim} H$ automorphism $\Rightarrow Ad_g(K)=K$ ie. $gKg^{-1}=K$).

例子: $G=D_8=\langle r,s: r^4=s^2=e, srs=r^{-1}\rangle$, $K=\langle r^2\rangle=Z(G)$, $H=\langle r^2,s\rangle\simeq C_2\times C_2$.

中心在自同构下不变, 故 K char G. 但H的3或元r^2, s, rs为二阶且交换, 故交换它们给出一自同构.

此自同构将 $K=\langle r^2\rangle$ 送到 $\langle s\rangle\neq K$. 则 K ~~char~~ H.